

University Placement Tracking Database

Oracle relational database design, SQL reporting and portfolio reconstruction

Project: Completed university database coursework reconstructed as a secure public portfolio case study.

Technologies: Oracle SQL, Oracle APEX, relational modelling, SQLite validation harness.

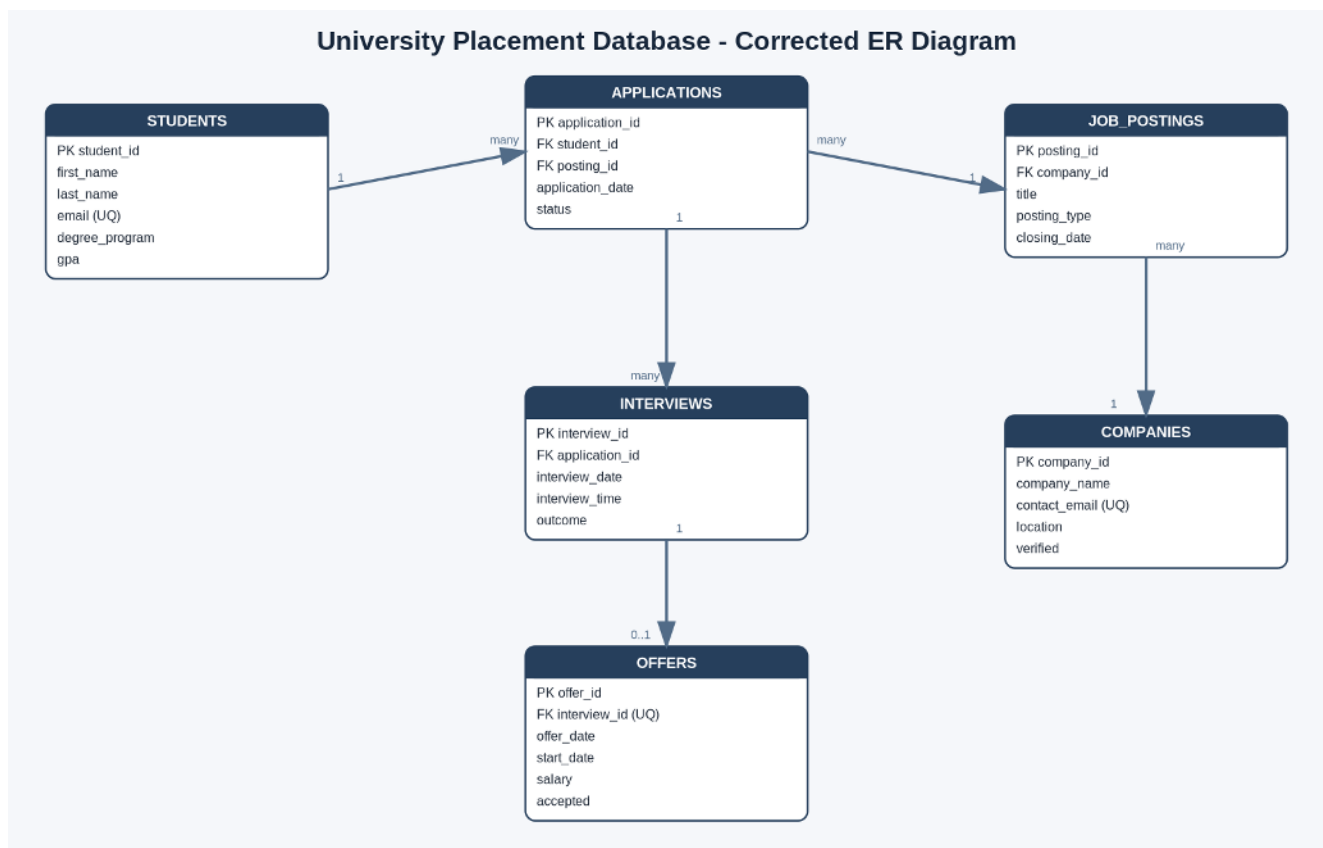
Students	Companies	Job postings	Applications	Interviews	Offers
5	3	4	7	4	2

Problem and Approach

The database supports a university placement office by connecting students, verified employers, opportunities, applications, interviews and offers. The reconstructed design stores each type of fact in one table and links records with keys, allowing operational reports without repeating names or company details throughout the database.

The public version removes exposed credentials, corrects the original ER diagram, strengthens constraints and replaces personal sample values with synthetic data.

Corrected Relational Model

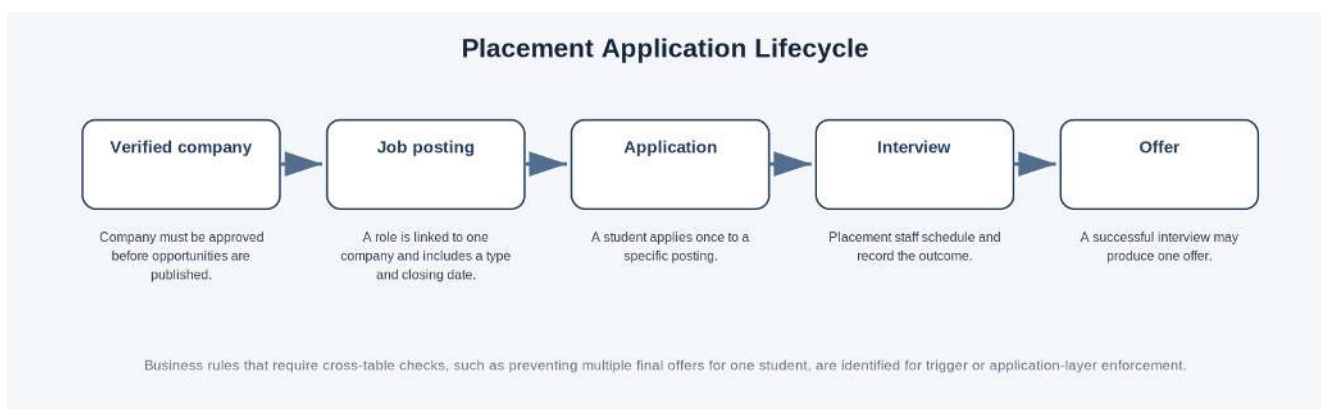


The corrected model restores the missing Interview entity, removes the duplicate Company entity and aligns relationships with the executable SQL.

Core Design Decisions

Decision	Reason
Separate entities	Prevents repeated student, company and posting information.
Composite application uniqueness	Blocks duplicate applications to the same opportunity.
Offer-interview uniqueness	Allows zero or one offer per interview.
Privacy-focused view	Supports routine reporting without exposing email or GPA.
Synthetic data	Keeps the public repository useful without publishing personal records.

Placement Lifecycle



The database supports the sequence from company verification and posting creation through application, interview and offer. Cross-table policies that need procedural enforcement are documented rather than overstated.

SQL and Integrity Controls

The Oracle scripts use primary keys, foreign keys, unique constraints, check constraints and indexes on principal join columns. Validation covers email uniqueness, GPA range, application status, interview outcome, offer acceptance, non-negative salary and chronological offer dates.

Reporting uses inner joins, a left join, aggregation, filtering, ordering, AVG and ROUND. A summary view deliberately excludes student email addresses and GPA values.

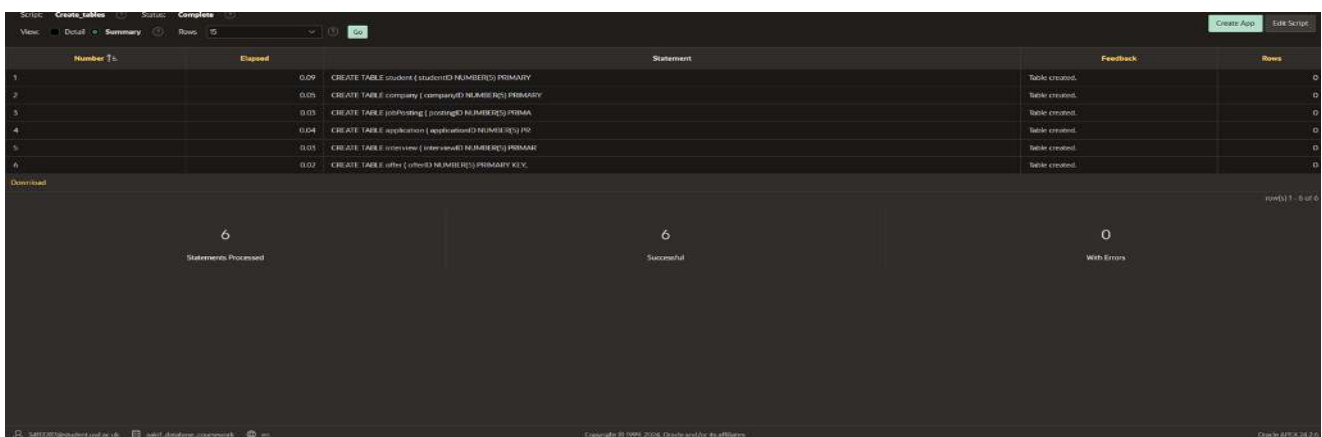
Validated Reporting Results

Metric	Validated result
Applications	7
Interviews	4
Accepted offers	2
Average GPA of placed students	3.38

Testing Strategy

Authentic APEX screenshots from the completed coursework show successful schema creation, data insertion and query execution. Because the university workspace is no longer available, an equivalent in-memory SQLite harness independently checks the reconstructed dataset and expected reporting values. The repository labels these two evidence types separately.

Original schema execution evidence



Number	Elapsed	Statement	Feedback	Rows
1	0.09	CREATE TABLE student (studentID NUMBER(5) PRIMARY	Table created.	0
2	0.05	CREATE TABLE company (companyID NUMBER(5) PRIMARY	Table created.	0
3	0.05	CREATE TABLE jobposting (postingID NUMBER(5) PRIMA	Table created.	0
4	0.04	CREATE TABLE application (applicationID NUMBER(5) PO	Table created.	0
5	0.05	CREATE TABLE interview (interviewID NUMBER(5) PRIMAR	Table created.	0
6	0.02	CREATE TABLE offer (offerID NUMBER(5) PRIMARY KEY,	Table created.	0

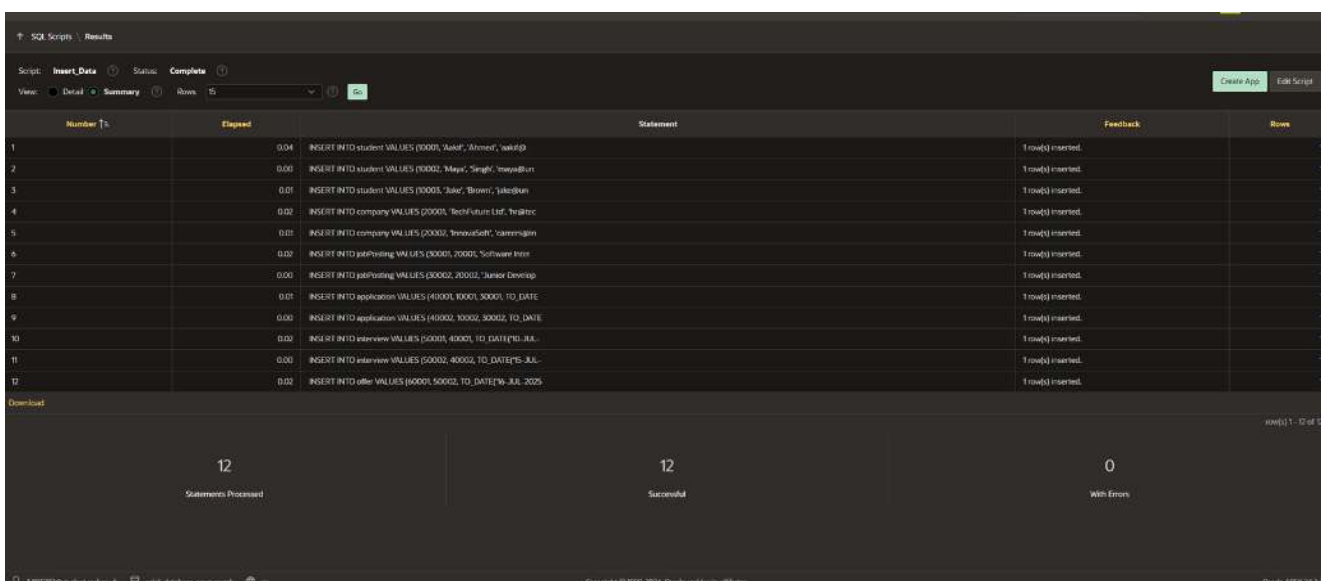
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6 Statements Processed

6 Successful

0 With Errors

Original sample-data execution evidence



Number	Elapsed	Statement	Feedback	Rows
1	0.04	INSERT INTO student VALUES (0000, 'Akaf', 'Ahmed', 'wakeup	1 row(s) inserted.	
2	0.00	INSERT INTO student VALUES (0000, 'Maya', 'Singh', 'mayasbu	1 row(s) inserted.	
3	0.01	INSERT INTO student VALUES (0000, 'Jade', 'Brown', 'jadedan	1 row(s) inserted.	
4	0.02	INSERT INTO company VALUES (0000, 'TechFuture Ltd', 'hr@tec	1 row(s) inserted.	
5	0.01	INSERT INTO company VALUES (0000, 'InnovateSoft', 'careers@in	1 row(s) inserted.	
6	0.02	INSERT INTO jobposting VALUES (0000, 2000, 'Software Engi	1 row(s) inserted.	
7	0.00	INSERT INTO jobposting VALUES (0000, 2000, 'Junior Develop	1 row(s) inserted.	
8	0.01	INSERT INTO application VALUES (4000, 0000, 0000, TO_DATE	1 row(s) inserted.	
9	0.00	INSERT INTO application VALUES (4000, 0000, 0000, TO_DATE	1 row(s) inserted.	
10	0.02	INSERT INTO interview VALUES (5000, 4000, TO_DATE('10-JUL-	1 row(s) inserted.	
11	0.00	INSERT INTO interview VALUES (5000, 4000, TO_DATE('15-JUL-	1 row(s) inserted.	
12	0.02	INSERT INTO offer VALUES (6000, 5000, TO_DATE('16-JUL-2025	1 row(s) inserted.	

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12 Statements Processed

12 Successful

0 With Errors

Security and Privacy

The original submission contained account information and a plaintext password. Those values were removed from every public asset. The portfolio also replaces personal sample data with fictional records and reserved example domains.

A production system would require role-based access, parameterised application queries, audit logging, retention rules and application- or trigger-level enforcement for policies that span several tables.

Corrections from the Original Submission

Issue	Portfolio correction
Duplicate Company and missing Interview in ERD	Rebuilt the diagram from the SQL and written rules.
Mixed naming conventions	Standardised plural table names and snake_case columns.
Weak nullability and validation	Added required foreign keys and domain checks.
Personal data in examples	Replaced with synthetic records.
Retired APEX workspace	Added offline validation without mislabelling it as Oracle evidence.

Skills Demonstrated

Relational modelling; Oracle SQL; entity relationship design; normalisation; constraints; joins; aggregate reporting; database views; test design; privacy-aware documentation; secure portfolio sanitisation.

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